



North South University

Course name : CSE 225
Section : 12
Semester : Summer 24
HW 02 Topic : Trees
Date : 02-10-2024

Submitted To :

Dr. Md. Ahsan Habib (AHbb)

Submitted by :

NAME : Mohammad Tushar Abdullah

ID : 2323766042

HOMEWORK 04 (part 01)

Question 01 : *Suppose you have a binary search tree (BST) with 31 nodes.*

What is the maximum possible height of the tree?

What is the minimum possible height of the tree?

Ans : For 31 nodes

maximum height is = $n - 1$

$$= 31 - 1$$

$$= 30$$

Minimum height = $\log_2(n+1) - 1$

$$= \log_2(31 + 1) - 1$$

$$= 5$$

Question 2: *You are given a binary search tree with a height of 5.*

What is the minimum number of nodes the tree can have?

What is the maximum number of nodes the tree can have?

Ans : For a binary search tree with a height of 5,

Minimum nodes = Height + 1

$$= 5 + 1$$

$$= 6$$

So, the minimum number of nodes with height 5 can have is 6.

The maximum number of nodes in a binary search tree with a height of 5 is N

Now , $N = 2^{h+1} - 1$ where $h = 5$ given,

$$\text{So, } N = 2^{5+1} - 1$$

$$= 64 - 1$$

$$= 63$$

Therefore, the maximum number of nodes the tree can have is 63.

Question 3: *If a binary search tree has 63 nodes,
what is the height of the tree when it is perfectly balanced?*

Ans : let $n = 63$;

& height of the perfectly balanced tree is h

$$\text{Then , } h = \log_2(n+1) - 1$$

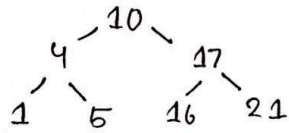
$$= \log_2(63 + 1) - 1$$

$$= 5$$

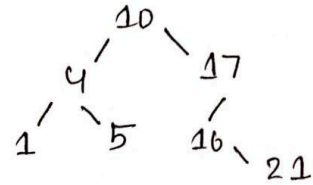
Therefore, the height of the tree when it is perfectly balanced is 5.

Ans to the question - 02

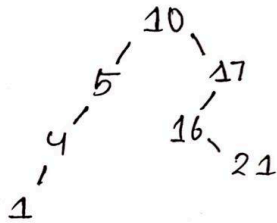
① Height - 02



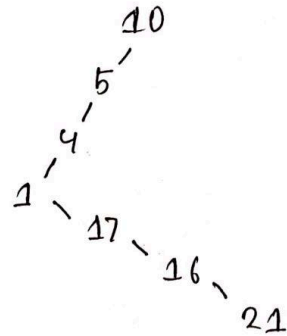
② Height - 03



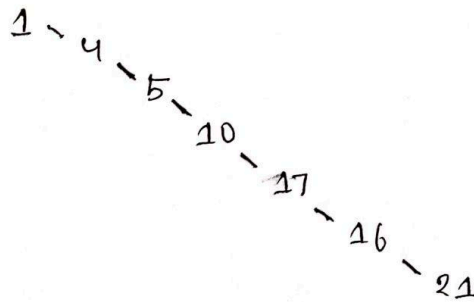
③ Height 04:



④ Height 05:



⑤ Height - 06



HOMEWORK 04 (part 02)

Ans to the question no : 01

In order : A, B, C, D, E, F, G, H, I

Ans to the question no : 02

Pre order : 25, 15, 10, 4, 12, 22, 18, 24, 50, 35, 31, 44, 70, 66, 90

Ans to the question no : 03

Pre order : F, B, A, D, C, E, G, I, H

Ans to the question no : 04

Postorder : A, C, E, D, B, H, I, G, F

Ans to the question no : 05

Post order : 4, 12, 10, 18, 24, 22, 15, 31, 44, 35, 66, 90, 70, 50, 25

